



Question Set

Question 1

A portfolio PnL is

$$\Pi = 4X - 2Y$$

with

$$\text{Var}(X) = 9, \quad \text{Var}(Y) = 16, \quad \text{Corr}(X, Y) = 0.35$$

Compute

$$\text{Var}(\Pi)$$

Question 2

Prove that independence implies zero covariance when second moments exist.

Question 3

Give a concrete example where covariance is zero but variables are dependent.

Question 4

Derive the minimum-variance hedge ratio by projecting portfolio PnL onto hedge PnL.

Question 5

State the law of large numbers and explain why it does not eliminate tail risk.

Question 6

Use the central limit theorem to approximate the distribution of the average of n independent daily returns with mean μ and variance σ^2 .

Question 7

A two-asset portfolio has weights 0.7 and 0.3, volatilities 20 percent and 35 percent, and correlation -0.25. Compute portfolio volatility.

Question 8

Explain why square-root-of-time VaR scaling can fail.

Question 9

Prove the variance decomposition

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$$

Question 10

What is convergence in distribution, and why is it weaker than convergence in probability?

Question 11

Explain Jensen's inequality and give a risk-management interpretation.

Question 12

A sample correlation matrix has a negative eigenvalue. What does this imply and how would you respond?